

Sam Thiel

Mathematician and Musician. sthiet@oberlin.edu. <https://stolenquotient2.github.io/>

EDUCATION:

Oberlin College, Oberlin, OH. Diploma expected June 2026.

Declared Majors in Mathematics and Musical Studies. Honors Candidate in Mathematics.

Deerfield Academy, Deerfield, MA. Diploma received May 2022.

RELEVANT COURSEWORK (Mathematics):

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| - Real Analysis (MATH 358) | - Group Theory (MATH 327) |
| - Topology (MATH 353) | - Foundations of Analysis (MATH 301) |
| - Probability (MATH 355) | - Computational Algebra (MATH 328) |
| - Harmonic Analysis (MATH 357) | - Linear Algebra (MATH 232) |
| - Mathematical Approaches to Music Theory (MATH 347) | - Introduction to Computer Science (CSCI 150) |

PROFESSIONAL EXPERIENCE:

Oberlin College and Conservatory, Oberlin, OH.

Research Assistant (with Chris Marx) (Aug. 2024–Present)

- Met weekly with Prof. Marx to work on his textbook on the intersection between Harmonic Analysis and Music.
- Composed in-text figures to help students visualize novel tools and results
- Developed online resources to enhance student learning by providing hands-on examples
- Designed end-of-chapter problems to provide essential practice in the textbook material.
- Proofread sections of the textbook for errors and clarity.

Teaching Assistant (For Susan Colley, Alexander Wilson) (Fall Semester 2024–Spring Semester 2025)

- Ran weekly 90-minute problem sessions each week for students taking Group Theory (MATH327).
- Collected, graded, and provided feedback on weekly homework assignments.
- Analyzed common mistakes and difficulties and maintained communication with the professor about these issues.
- Developed individual rapport with students in order to build a safe and supportive learning environment.

Homework Grader (For Susan Colley) (Fall Semester 2022–Spring Semester 2024)

- Collected and graded weekly homework for students taking Linear Algebra (MATH 232, proof-based) and Multivariable Calculus (MATH231, non-proof based).
- Provided insightful comments and feedback to promote student learning.
- Analyzed common mistakes among students and relayed information back to the professor

Research:

- **Partial Results of the HRT Conjecture**, *Research Experience for Undergraduates, UWashingon–Seattle, June–July 2025.*

Presentations:

- **Costello Divisibility: Explorations of a Comedic Division Algorithm**, *2024-2025 Oberlin Undergraduate Research Symposium, Apr. 2025.*

Publications:

- **Costello Divisibility: Explorations of a Comedic Division Algorithm** (with Alexander N. Wilson). *The College Mathematics Journal*, submitted.

Skills & Software:

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| • Microsoft Office Suite | • Python | • Ableton Live |
| • Google Suite | • GitHub | • MuseScore |
| • LaTeX/TexShop | • Affinity Designer | • Logic Pro |